Explainable AI-based Crop Disease Detection System using Leaf Image Analysis

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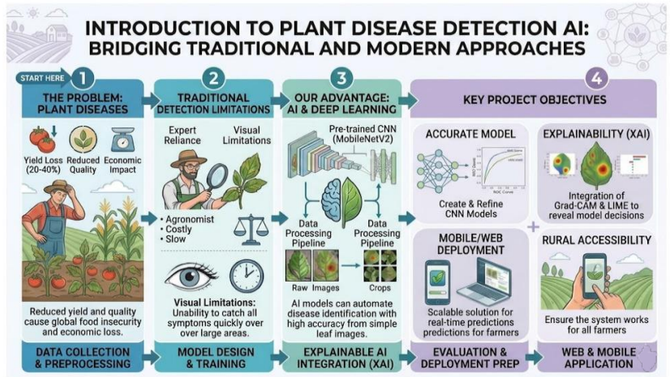
INTRODUCTION

Plant diseases significantly reduce crop yield and quality, causing major economic losses for farmers worldwide. This project uses deep learning, especially CNN-based models, to automatically detect plant diseases from leaf images with high accuracy. The system aims to be deployed as a web and mobile application, enabling farmers to get real-time predictions and make informed decisions.



PROBLEM STATEMENT

Plant diseases significantly impact agricultural productivity, causing major yield losses and economic damage. Traditional detection methods depend on expert knowledge, making them costly, slow, and inaccessible to many farmers. Although deep learning models provide high accuracy, they often lack explainability and real-world adaptability, highlighting the need for scalable and practical solutions.

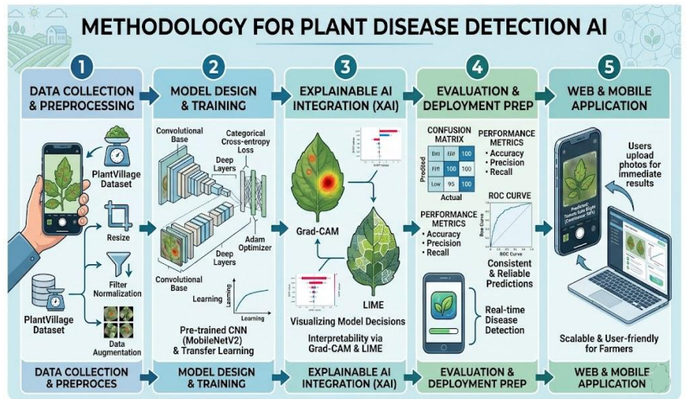
OBJECTIVES

The project aims to develop a CNN-based model for accurate plant disease classification and evaluate its performance using key metrics. It integrates explainable AI techniques like Grad-CAM, LIME, and SHAP to improve transparency. The system is designed for scalable deployment as a web/mobile app, enabling farmers to get real-time predictions. It also focuses on making the solution accessible to farmers in rural and semi-urban areas with limited resources. Overall, the goal is to create a cost-effective, reliable, and user-friendly system for practical agricultural use.

METHODOLOGY

The methodology of this project involves collecting and preprocessing plant leaf images from the PlantVillage dataset, including resizing, normalization, and augmentation to improve model performance. A CNN-based multilayer perceptron model is then designed and trained to classify different plant diseases accurately.

The model is evaluated using performance metrics such as accuracy, precision, recall, and ROC curves to ensure reliability. Explainable AI techniques like Grad-CAM, LIME, and SHAP are integrated to interpret model predictions and enhance trust. Finally, the trained model is prepared for deployment as a scalable web/mobile application for real-time disease detection.



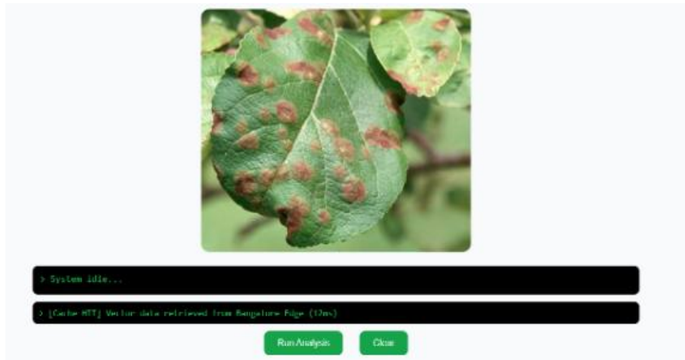
IMPLEMENTATION

The system is implemented using a pre-trained CNN model (MobileNetV2) with transfer learning for accurate disease classification. Image preprocessing techniques like resizing, normalization, and augmentation are applied to improve performance. Explainable AI methods such as Grad-CAM and LIME are integrated to visualize model decisions. A web application is developed using React.js and Node.js to allow users to upload images and receive real-time predictions with confidence scores. The backend processes images through a trained model and generates predictions efficiently. The system is designed to be scalable and accessible for real-world agricultural use, especially for farmers.

RESULTS

The system achieved high accuracy in plant disease classification using the CNN model, evaluated through metrics like accuracy, precision, and recall.

Explainable AI techniques provided clear visual insights, and the web/mobile interface enabled real-time, practical disease detection for farmers.



CONCLUSION AND FUTURE SCOPE

The project successfully developed an accurate and explainable plant disease detection system using CNN and XAI techniques. It provides real-time predictions through a user-friendly web/mobile interface, making it practical for farmers.

In future, the system can be improved by using real-field datasets, adding more crop varieties, and enhancing mobile deployment for wider accessibility.

REFERENCES

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